

SSH: Connecting from Linux to Linux using OpenSSH Server

Assignment study PDF — English

1. What is SSH?

Secure Shell (SSH) provides encrypted remote administration of computers. Linux-to-Linux administration commonly uses OpenSSH, which provides the ssh client and sshd server.

2. Install OpenSSH Server

```
On Ubuntu/Debian: sudo apt update  
  
sudo apt install openssh-server  
  
sudo systemctl enable --now ssh
```

3. Find the server address

Run `ip addr` on the server and identify its reachable IPv4 address. The client then connects using `ssh username@SERVER_IP`.

4. First connection

From the Linux client run: `ssh username@SERVER_IP`. The first connection may ask you to verify the server host key. Enter the user's password when password authentication is enabled.

5. Key-based authentication

SSH keys use a private key on the client and a public key authorized on the server. A common command is `ssh-keygen -t ed25519`. The public key can be installed using `ssh-copy-id username@SERVER_IP`.

6. Configuration

The SSH server configuration is normally stored in `/etc/ssh/sshd_config`. After changes, validate the configuration and restart/reload the SSH service carefully.

7. Security

Use strong authentication, restrict unnecessary access, keep OpenSSH updated, disable direct root login when appropriate, and prefer key-based authentication for administrative access.